EchoForge: A Strict-Open Synthetic Benchmark for Low-Altitude UAS Radar and Multimodal Sensing

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***Abstract*—EchoForge is a strict-open synthetic benchmark for low-altitude uncrewed aerial system (UAS) radar and multimodal sensing research. Strict-open means that assumptions, code, configuration, detector-view schemas, and paper evidence are reviewable from public or generated sources, while generated arrays and measured raw files remain outside Git. The paper-facing positive class is a fixed-wing pusher-prop public proxy with explicit uncertainty; it is not a measured-platform claim and is not a fielded-sensor surrogate. The benchmark run contains 10,000 scenario groups, 50 positive groups, and 30,000 phase records, with a blind group-locked holdout scored once after train/CV selection lock. Primary KPI: LCB95 Recall@≤1%FPR. The accepted human-engineered prior fusion baseline is the comparator. On the 4,500-record blind holdout, with 24 positive records from eight positive scenario groups, the Engineered Intelligence (EI) candidate reaches point Recall@≤1%FPR of 0.833 vs 0.292 (+54.1 pp, +185%, 2.85x) and LCB95 Recall@≤1%FPR of 0.699 vs 0.083 (+61.6 pp, +742%, 8.42x). EI also raises average precision from 0.128 to 0.825 (+0.697, about +545%) and cuts selected-threshold false positives from 49 to 1 (98.0% fewer false alarms). EI ROC AUC is 0.917, below prior fusion's 0.938, so the claim is improved low-FPR operating behavior for this synthetic public-proxy benchmark, not universal rank dominance. These results should be read as reproducible benchmark evidence under declared assumptions, not as measured truth, named-platform equivalence, or operational performance.**

***Index Terms*—Synthetic radar, UAS detection, public proxy, micro-Doppler, calibration, reproducibility, compare-only measured anchor.**

I. Claim Boundary and Contributions

EchoForge is intentionally narrower than a measured radar collection. It publishes public-proxy signatures and detector-facing products with explicit uncertainty and validation tiers, while keeping generated arrays, solver outputs, and measured raw files out of Git. The radar and detection framing follows standard signal-processing and CFAR practice rather than a sensor-surrogate claim [1]–[3]. The paper uses repository-compatible legacy identifiers only where required for tooling; the paper-facing positive class is named *fixed-wing pusher-prop public proxy*. No measured-platform truth, proprietary-equivalent behavior, or classified fidelity is claimed for any object, platform, or sensor.

The paper has one main experiment: compare accepted human-engineered radar/cue baselines and prior fusion against a locked EI sparse late-fusion candidate on the same blind, group-locked synthetic holdout. The contributions are:

- a strict-open radar benchmark lane with explicit waveform, range-Doppler, clutter/noise/RFI, acoustic, passive-RF, and detector-view assumptions;
- a rare-positive, low-false-alarm evaluation protocol centered on LCB95 Recall@≤1%FPR, with AP, ROC AUC, calibration, and false-alarm families as guardrails;
- a locked EI candidate that searches only train/CV detector-view scores, then applies sparse nonnegative fusion and odds calibration before one blind-holdout pass;
- appendix evidence that explains how the fixed-wing pusher-prop public proxy, bird/RC hard negatives, sensor archetypes, and clutter/noise/RFI stressors are modeled without making measured-platform claims.

A. Core Experiment

The core experiment compares classical radar and cue detector branches, tabular and sequence ML baselines, an accepted human-engineered prior fusion baseline, and an EI sparse calibrated late-fusion candidate. Scenario groups are locked before phase expansion, train/CV rows are used for candidate discovery and calibration, and the blind holdout is scored once after selection lock. The benchmark output remains synthetic public-proxy evidence only.

The claim tested is deliberately narrow: under the declared synthetic public-proxy generator and group-locked split, the locked EI sparse late-fusion candidate improves the low-FPR operating point over the accepted human-engineered prior fusion baseline. This comparator is not a straw baseline; it is the strongest non-EI late-fusion lane assembled from radar/cue branches under the same split and train/CV thresholding policy. The claim is not fielded-sensor equivalence, not a measured Iranian-drone signature claim, and not an operational false-alarm-rate claim.

Benchmark positioning is deliberately compare-only. KTH provides a public 77 GHz FMCW measured anchor for drone/bird/human observable-shape checks; Open Radar and RAD-DAR/RDRD provide public radar and range-Doppler reference points; and public UAS micro-Doppler studies bound the observable families used for small aerial objects [4]–[10]. EchoForge does not replace those measured collections. It provides a strict-open synthetic lane for controlled hard negatives, leakage checks, and low-FPR detector comparison when measured positive-class truth is not available.

Figure 1 summarizes the signal chain and evidence boundary.

II. Data Processing and Evidence Flow

Figure 2 is the reviewer-facing processing trace. EchoForge first samples scenario groups from public-proxy strata, then

TABLE I
CORE EXPERIMENT ROADMAP

Lane	What is modeled	Why it is in the paper	Reader-facing conclusion
Sensor branches	X/Ku radar, S-band radar, GBAD cueing, acoustic cueing, and passive-RF provenance	Shows the component evidence available before fusion	Branches are sanity checks and failure-slice baselines
Prior ML	Tabular and sequence learners trained only on detector-facing features	Tests ordinary learned baselines under the same split	Useful controls, not the accepted-practice comparator
Accepted fusion	Human-engineered layered fusion with train/CV thresholding	Represents the accepted human-engineered prior fusion baseline	Main comparator for EI
EI sparse fusion	Train/CV candidate discovery, sparse non-negative component fusion, monotone odds calibration, and one locked holdout pass	Tests whether EI improves the low-FPR operating point	Main challenger; claims are limited to this synthetic public-proxy benchmark

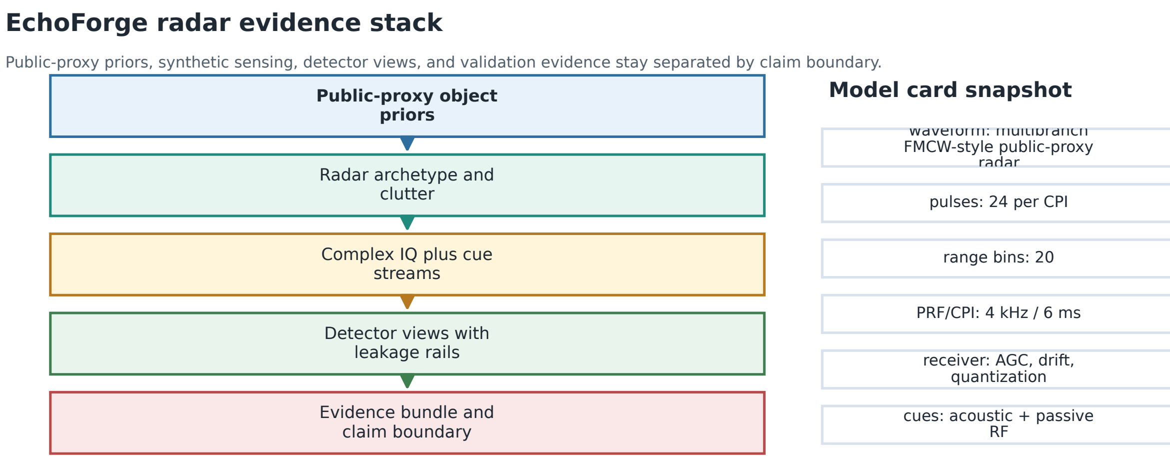

Fig. 1. Radar signal chain and claim boundary. Public-proxy object priors, radar archetype assumptions, clutter and artifact priors, detector views, and evidence outputs are separated so the benchmark can be audited without conflating synthetic products with measured truth. Detector outputs feed both the accepted prior fusion baseline and the EI sparse fusion candidate.

expands each group into phase records, generates synthetic IQ and cue products, materializes detector-view features through a denylisted schema, locks EI discovery and calibration on train/CV rows, and scores the blind holdout once. The important boundary is early: group IDs, split keys, labels, time locks, and audit headers can be inspected by validation code but are denied to model-facing features.

III. RADAR AND MULTIMODAL GENERATIVE MODEL

EchoForge uses a clean-room signal-chain proxy rather than a proprietary sensor surrogate. The received-power proxy is written as

$$P_r \propto \frac{P_t G_t G_r \lambda^2 \sigma(\theta, f, \phi)}{(4\pi)^3 R^4 L}, \tag{1}$$

where the radar cross section σ is an aspect- and frequency-dependent public-proxy quantity, R is slant range, and L is an aggregate loss term. Complex IQ is generated as a phase-stable target term plus clutter, interference, and receiver impairment terms,

$$x_{m,n} = a_{m,n} e^{j(2\pi f_d m T_r + \varphi_n)} + c_{m,n} + \epsilon_{m,n}, \tag{2}$$

with m indexing pulses and n indexing range bins. The benchmark keeps these values synthetic: the absolute scale is a proxy, and the public-proxy object family is defined by priors rather than measured signatures, as in conventional micro-Doppler and small-UAS radar modeling [8]–[13].

The range-Doppler proxy uses FMCW/chirp wording for the synthetic range product and pulse/CPI wording for Doppler indexing. The derived quantities are

$$\Delta R = \frac{c}{2B}, \qquad \Delta f_D = \frac{1}{T_{\mathrm{CPI}}}, \qquad \Delta v = \frac{\lambda}{2T_{\mathrm{CPI}}}, \tag{3}$$

with approximate unambiguous velocity $\lambda f_{\mathrm{PRF}}/4$ for the simplified branch cards. Target priors include an aspect-dependent RCS envelope, Swerling-like fluctuation proxy, 150–217 Hz

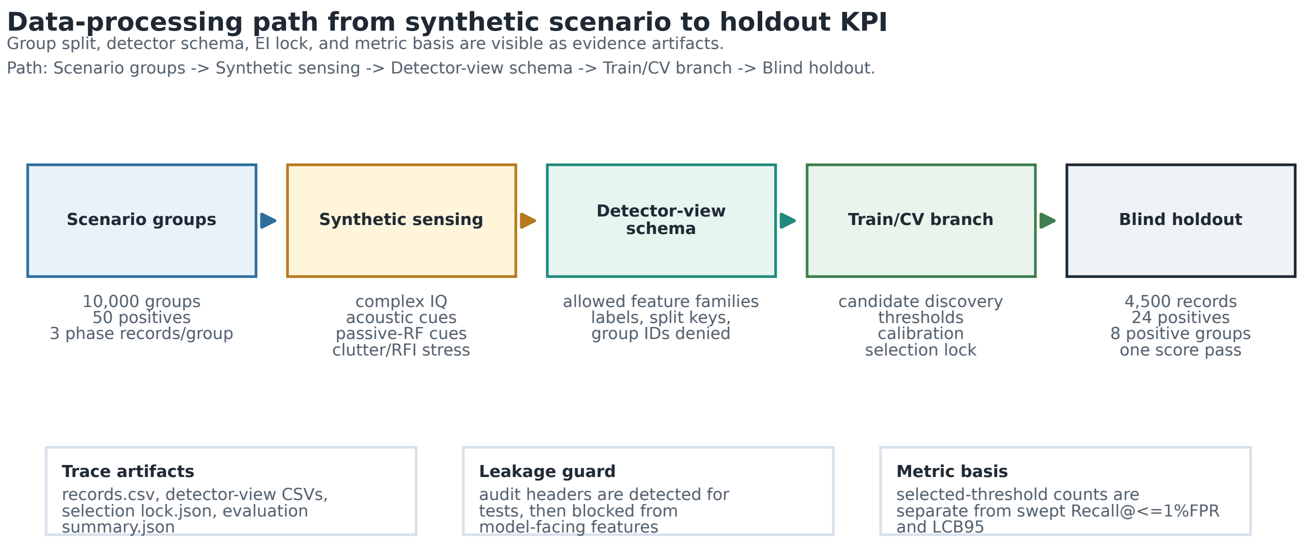

Fig. 2. End-to-end data-processing path. Scenario groups and phase records generate synthetic radar/cue products; detector-view materialization enforces feature denylisting; train/CV rows drive candidate discovery, thresholds, and monotone calibration; and the blind holdout is scored once after selection lock. The figure is a traceability diagram, not an additional performance claim.

TABLE II
DATA PROCESSING TRACEABILITY CHECKLIST

Stage	Produced artifact	Reviewer check	Failure mode guarded
Scenario sampling	`scenario_manifest.csv`, `dataset_manifest.json`	group IDs and split roles exist before phase expansion	row-level leakage through repeated phases
Synthetic sensing	`records.csv`, raw IQ references, acoustic and passive-RF cue summaries	generated radar/cue products inherit public-proxy priors and group split	undocumented signal-source mixing
Detector views	detector-view CSVs and `detector_view_schema_evi`…	labels, group IDs, split keys, audit headers, and generator metadata are blocked	shortcut learning from audit fields
Train/CV branch	`selection_lock.json`, component weights, thresholds, calibrators	candidate search, weighting, thresholding, and calibration are train/CV-only	holdout tuning or silent reselection
Blind holdout	`evaluation_summary.json`, `main_kpi_gain_table.csv`	selected-threshold counts, swept Recall@$\leq$1%FPR, and LCB95 are separate fields	conflating locked-threshold counts with swept low-FPR recall
Paper evidence	`paper_evidence_manifest.`…, `data_processing_trace_rows`…	every headline figure/table is downstream of generated evidence	hand-edited result drift

pusher-prop micro-Doppler proxy, and hard-negative bird/RC cadence bands [8], [9], [11], [12], [14]. Clutter/noise/RFI priors cover Weibull and K-like clutter, sea/terrain/weather stress, AGC compression, clock drift, quantization, dropped CPI, PRF ambiguity, Doppler folding, calibration offset, multipath masking, and RFI burst regimes [15]–[17]. Table III is the compact model card used by the evidence bundle. The values are public-proxy assumptions, not claims about any operational sensor.

The receiver impairment set includes AGC compression, clock drift, quantization, dropped CPIs, PRF ambiguity, Doppler folding, calibration offset, and multipath masking. Acoustic cues are modeled as node-wise spectral cadence and inter-node agreement; passive-RF cues are modeled as no-signal, RFI burst, clock-offset, and provenance-quality geometry. These definitions are public-proxy abstractions, not measured claims, and are aligned with the literature on measured-anchor comparison rather than truth transfer [4]–[7].

IV. PUBLIC-SOURCE DETECTOR ARCHETYPES

EchoForge detector branches are public-source archetype envelopes. The public examples are used to bound branch roles and observable families, not to reproduce vendor processing. The X/Ku C-UAS branch is anchored by public descriptions of low-slow-small radar behavior such as Blighter A400 and KuRFS [18], [19]. The tactical S-band branch follows public

TABLE III
RADAR MODEL CARD AND DERIVED RESOLUTION ASSUMPTIONS

Branch	f_0 GHz	BW MHz	PRF/CRF Hz	CPI ms	Chirps	Span m	ΔR m	Δv m/s
X/Ku C-UAS	10.0	600	4000/4000	6.0	24	5.0	0.250	2.498
S-band AESA/MHR	3.1	180	4000/4000	6.0	24	16.7	0.833	8.059
GBAD 3D/4D	10.3	300	4000/4000	6.0	24	10.0	0.500	2.426

MHR/nMHR-style hemispheric surveillance roles [20]. The GBAD cueing branch follows public wide-area 3D/4D radar roles such as Giraffe 1X and Sentinel summaries [21], [22]. In all cases, role-compatible public envelope means plausible branch role and observable family, not sensitivity, ECCM, proprietary clutter rejection, exact Pd/Pfa, or military performance.

V. Scenario Design and Detector Views

The main run consists of 10,000 scenario groups, 50 positive groups, and 30,000 phase records generated from a single seed. Each group expands into three locked windows: initial take-up, climb transition, and cruise altitude. Split assignment occurs at the scenario-group level so all phase records from one group share the same split role and fold. In the v2 split, the default holdout contains 1,500 groups with 8 positive groups, leaving 42 positive groups for train/CV; every train/CV fold receives positives, and the default benchmark remains group-random holdout. The paper evidence also supports a targeted holdout policy for diagnostic out-of-distribution slices.

Figure 3 shows the group-locked split and the leakage diagnostics that accompany it. The paper evidence records label-split counts, site/range/aspect/noise counts, hard-negative family counts, marginal imbalance scores, a stratum-only baseline, a metadata-only baseline, a label-shuffle sanity test, a forbidden-feature canary test, and a nearest-neighbor train/holdout similarity scan.

The detector-facing views remain intentionally narrower than the generator state. The high-resolution X/Ku, tactical S-band, and GBAD branches expose radar-specific numeric summaries; the acoustic branch exposes cue statistics; the passive-RF branch exposes no-signal and provenance-quality geometry; and the fusion branch exposes calibrated late-fusion scores. The evidence bundle now emits a detector-view schema table listing allowed feature families and audit-only fields. Restricting the model view prevents labels, split keys, group identifiers, and hidden generator metadata from becoming shortcuts.

The hard-negative roster is regionalized for Gulf/coastal site archetypes. Bird stressors are split into single bird, flock aggregate, shorebird/wader, gull/tern, raptor/falcon, flamingo or other large bird, seabird/cormorant, and seasonal migratory density families. The taxonomy uses public biodiversity context for regional prevalence and published drone/bird micro-Doppler behavior for wingbeat and spectral-overlap assumptions [8], [9], [23]. RC fixed-wing aircraft are modeled as a separate benign hard-negative family: hobby-class airframes, prop cadence overlap, short-to-medium range, variable maneuvering, and speed/aspect overlap with the positive public proxy. They are not target proxies.

VI. Evaluation Protocol and Calibration

Let s_i be the detector score for record i. For the EI candidate, the fused score is a nonnegative weighted sum over component scores,

$$z_i = \sum_{j=1}^{k} w_j s_{ij}, \qquad w_j \geq 0, \qquad \sum_{j=1}^{k} w_j = 1. \tag{4}$$

The paper-facing calibration step is a geodesic-odds transform implemented as a monotone log-odds map,

$$\hat{p}_i = \sigma\left(a \log \frac{z_i + \epsilon}{1 - z_i + \epsilon} + b\right), \tag{5}$$

where σ is the logistic function and ϵ prevents boundary singularities. We use train/CV rows only to fit thresholds and calibrators, and we score holdout rows once after the selection lock is written.

The main metrics are ROC AUC, average precision, fixed-FPR recall, accuracy, precision, recall, false-positive rate, F1, Brier score, and expected calibration error (ECE). Group-block bootstrap confidence intervals are computed by resampling scenario groups, not individual phase rows, so that uncertainty reflects the locked group structure rather than an artificially inflated sample size [24]–[27]. The paper evidence bundle also records PR and ROC curve traces and calibration bins, which matches standard classifier-evaluation and calibration practice [28]–[35].

A. Primary KPI

The headline KPI is the lower 95% group-block bootstrap bound of recall at FPR $\leq$ 1%, abbreviated *LCB95 Recall@≤1%FPR*. That choice is deliberate: the benchmark is rare-positive, false-alarm-constrained, and group-correlated, so the raw point estimate of recall is useful but not sufficient as the primary decision metric. AP, ROC AUC, raw recall, precision, F1, calibration, and false-alarm breakdowns remain diagnostic guardrails rather than the headline KPI.

Three operating quantities are kept separate. Selected-threshold TP/FP/FN counts use the threshold fixed by train/CV or by the EI selection lock. Recall@≤1%FPR is a swept holdout operating-point diagnostic: among thresholds satisfying the 1% FPR cap, it reports the best recall. LCB95 is the 2.5th percentile of the group-block bootstrap distribution for that swept low-FPR recall, with scenario groups as bootstrap units.

Figure 4 shows the primary KPI first and keeps AP, ROC/PR curves, and calibration as supporting diagnostics. Table VII is

TABLE IV
PUBLIC DETECTOR ARCHETYPE CARDS

Branch	Public role envelope	Detector-visible products	Excluded truth fields
X/Ku C-UAS	Low-slow-small search, high-resolution range-Doppler, bird/RC discrimination	Range-Doppler concentration, micro-Doppler spread, clutter stress	Classified sensitivity, proprietary clutter maps, exact Pd/Pfa
Tactical S-band AESA/MHR	Hemispheric surveillance, track-while-scan, medium tactical cueing	Coarser velocity bins, track confidence, multipath stress	Vendor processing, ECCM, deployment geometry
GBAD 3D/4D	Wide-area cueing, revisit delay, radar-horizon stress	Cue confidence, stale-track state, horizon masking	Engagement-quality behavior, operator workflows
Acoustic network	Spectral cadence and cross-node agreement	Propulsion cadence, bearing/time consistency, weather noise	Exact microphone layout, field localization performance
Passive RF	No-signal behavior, RFI bursts, provenance quality	Missingness, sparse RF cues, RFI/provenance flags	Emitter libraries, operator identity, payload commands
Fusion C2	Source confidence, cross-sensor confirmation, false-track accounting	Late-fusion score, source quality, stale-track penalty	Operational C2 logic or deployment topology

TABLE V
VALIDATION TIERS AND CURRENT STATUS

Tier	Status	Boundary
Synthetic corpus generation	pass	Scenario groups, detector views, and raw products are generated from a reproducible seed
Detector split isolation	pass	Holdout rows are excluded from calibration, thresholding, and candidate selection
Compare-only measured anchor	pass	KTH is used only for distribution-shape checks and hard-negative realism
Generative-origin self-classification	pass	Manifest-based LOC audit reports an auditable self-classification, not authorship proof
Named-platform truth claim	prohibited	No measured-object equivalence or operational performance claim is made

TABLE VI
REGIONAL HARD-NEGATIVE FAMILIES

Family	Proxy band	Main confusion mechanism
Shorebird/wader	5–14 Hz wingbeat	Small RCS, dense migration, low coastal returns
Gull/tern	2.5–8 Hz wingbeat	Moderate body returns and coastal flocking
Raptor/falcon	1.5–6 Hz wingbeat	Fast individual tracks and glide/flap transitions
Flamingo/large bird	1.2–4.5 Hz wingbeat	Larger body return, slower wingbeat, wetland corridors
Seabird/cormorant	2–7 Hz wingbeat	Sea-clutter interaction and low marine tracks
RC fixed-wing	40–220 Hz prop cadence	Prop cadence, speed, and aspect overlap with positives

the paper's main result ledger, and Table VIII keeps the raw holdout counts visible. In the v2 run, the holdout split contains 4,500 records, of which only 24 records from 8 positive scenario groups are positive. The EI candidate improves AP, selected-threshold precision, false-positive control, and swept Recall@$\leq$1%FPR over the accepted human-engineered prior fusion baseline. Its ROC AUC is lower than the prior fusion baseline, so the result should be read as better low-FPR operating behavior for this benchmark rather than universal rank dominance. Because the positive holdout has $n = 8$ groups, phase and family conclusions are diagnostic.

The gain row is the direct table interpretation: EI raises the primary LCB95 metric by +0.616, raises point Recall@$\leq$1%FPR by +0.541, and reduces selected-threshold false positives from 49 to 1 while conceding ROC AUC.

The evidence bundle also records the canary forbidden-feature test, the label-shuffle sanity test, the stratum-only baseline, the metadata-only baseline, and the nearest-neighbor train/holdout scan. Those tests are not leaderboard scores; they are leakage and realism diagnostics that make the benchmark harder to overfit.

TABLE VII
MAIN KPI GAIN LEDGER VERSUS ACCEPTED PRIOR FUSION

Metric	Prior	EI	Change
LCB95 Recall@$\leq$1%FPR	0.083	**0.699**	**+742%, 8.42x**
Point Recall@$\leq$1%FPR	0.292	**0.833**	**+185%, 2.85x**
Average precision	0.128	**0.825**	**about +545%**
Selected-threshold FP	49	**1**	**98.0% fewer**
F1	0.241	**0.837**	**+247%**
ROC AUC	0.938	**0.917**	**guardrail trade-off**

Figure 5 highlights that the initial take-up window remains

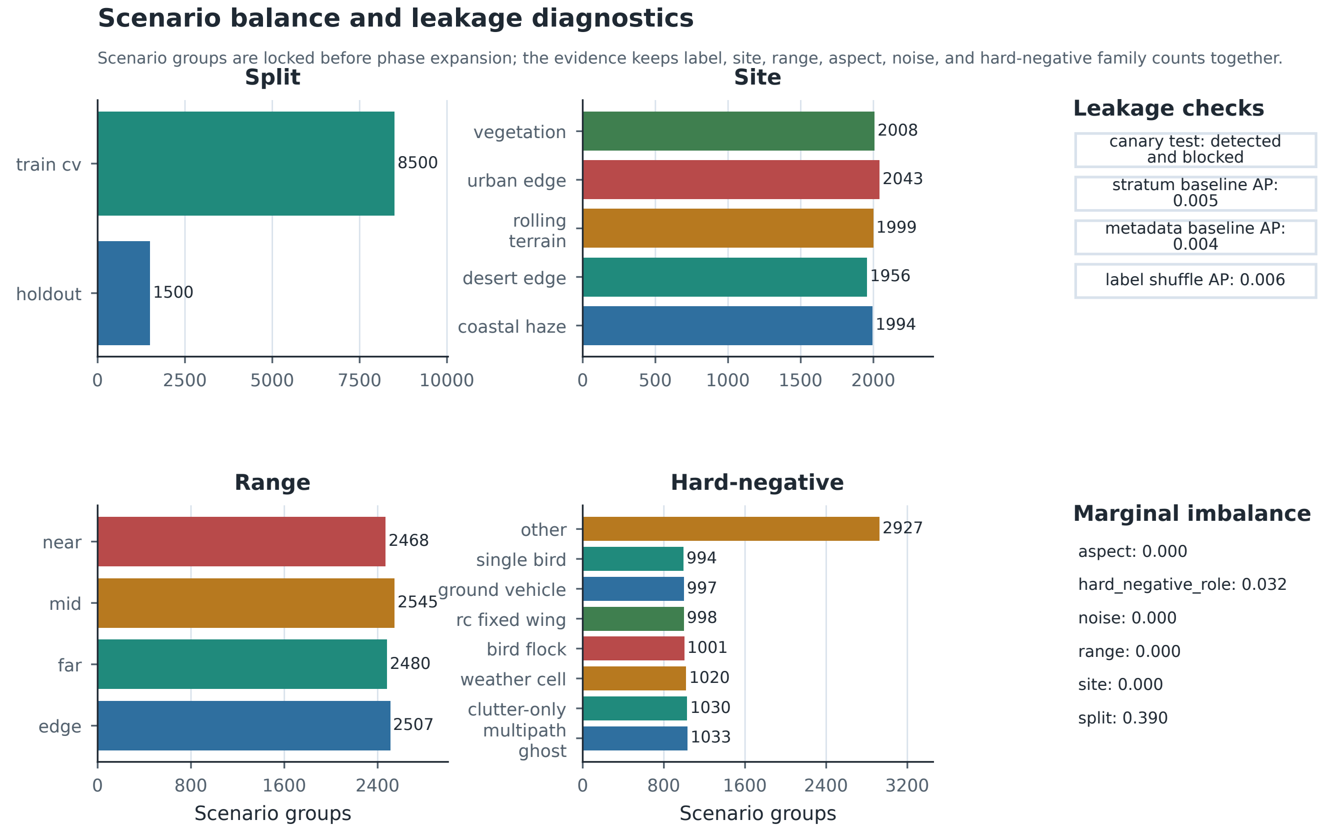

Fig. 3. Scenario balance and leakage diagnostics. Scenario groups are locked before phase expansion, and the evidence bundle keeps split, label, site, range, aspect, noise, and hard-negative family counts together with canary and sanity checks. Audit-only split/group headers are reported as detected and blocked by the model-facing schema rather than as detector features.

TABLE VIII
HOLDOUT SUMMARY SEPARATING RANKING, SELECTED-THRESHOLD COUNTS, AND SWEPT FIXED-FPR METRICS

Method	AP	ROC AUC	F1	ECE	Threshold source	TP/FP/FN	Sel. FPR	Recall@$\leq$1%FPR	LCB95
accepted prior fusion	0.128	0.938	0.241	0.016	train/CV max-F1	10/49/14	0.010947	0.292	0.083
EI candidate	0.825	0.917	0.837	0.002	EI selection lock	18/1/6	0.000223	0.833	0.699
EI gain vs prior	+0.697	-0.021	+0.596	-0.014	locked vs train/CV	+8/-48/-8	-0.010724	+0.541	+0.616

the hardest phase, while the cruise window is the most separable. The cross-method false-positive burden shows whether the strongest methods overreact to birds, RC fixed-wing aircraft, weather, clutter-only counterfactuals, or other hard-negative families. The point of the breakdown is not to optimize evasion; it is to expose robustness failure modes.

VII. Engineered Intelligence

The advanced lane is reported as engineered intelligence rather than as an opaque model handle. The reader-facing story is train/CV-only candidate discovery, evolved feature families, sparse calibrated fusion, and a single blind holdout score pass after the lock. The evidence bundle keeps the raw component handles and search traces in appendix artifacts, while the paper foregrounds the stage summaries that matter for review.

Mechanically, EI uses train/CV-only score search to choose component scores, sparse nonnegative late fusion to keep the lock interpretable, geodesic-odds calibration for monotone probability mapping, and a written selection lock before the single blind holdout score. The component weights and modality/control tables below are interpretability controls rather than separate main-claim evidence.

This distinction matters because a post-hoc view control can be strong on one metric without being the final EI artifact. In this run, the passive-RF-only view has higher AP and swept Recall@$\leq$1%FPR than the full EI candidate, which is an important diagnostic red flag rather than a contradiction to hide. Its selected-threshold F1 and ECE are worse, and it does not carry the same calibrated-fusion selection lock. We therefore report EI as the pre-specified, calibrated selected-threshold artifact, not as a universal dominator of every view control.

Figure 6 shows the reader-facing EI workflow, while Figure 7 replaces an opaque winner label with a transparent review surface. The component weights are reader-facing summaries of the EI sparse fusion, while the ab-

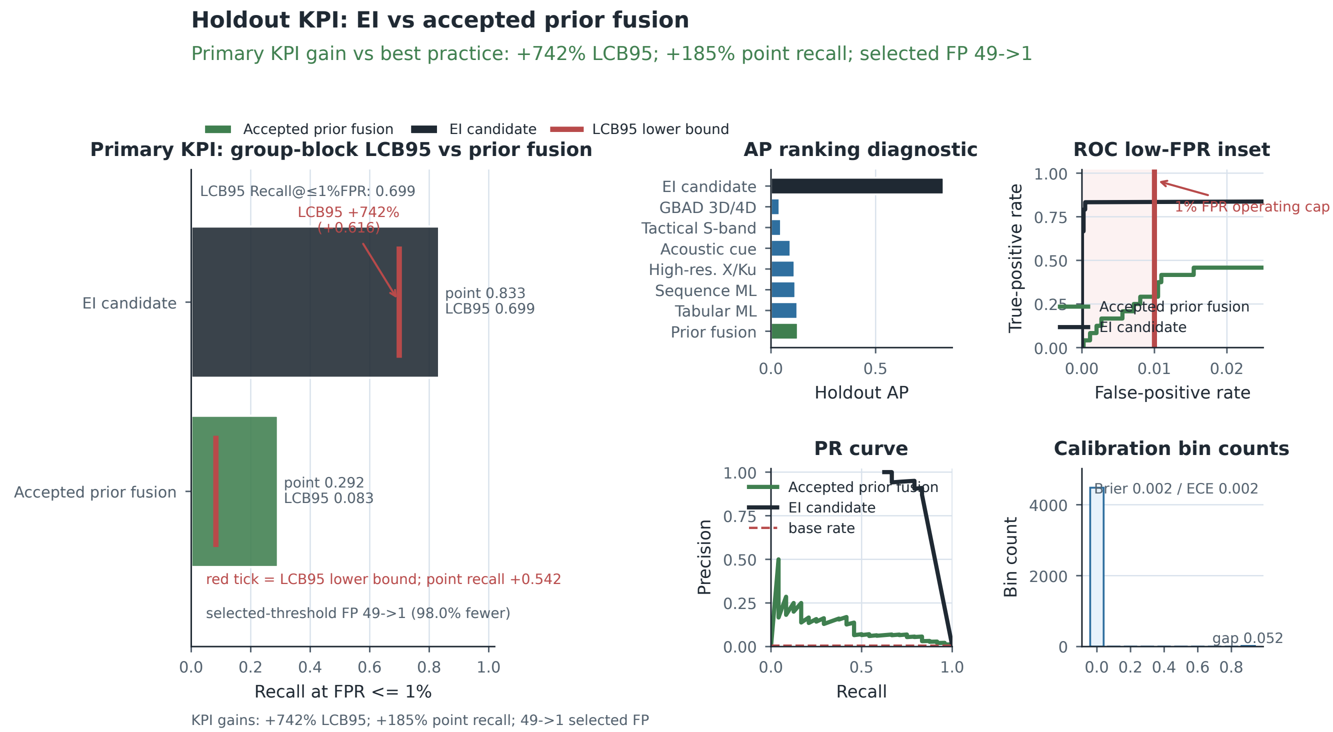

Fig. 4. Holdout primary KPI, ranking diagnostics, low-FPR ROC/PR behavior, and calibration. Red vertical ticks are group-block LCB95 lower bounds, and the pale red shaded zone with the thick red line marks the 1% FPR operating cap. Recall@≤1%FPR and its LCB95 are the headline quantities; AP, ROC AUC, PR shape, and calibration bins are supporting diagnostics.

TABLE IX
EI CANDIDATE DISCOVERY AND LOCKED SCORING ALGORITHM

Step	Operation
1	Read train/CV detector-view scores after the denylist removes labels, split fields, group IDs, time locks, and audit headers.
2	Generate candidate component families from radar, passive-RF, acoustic, transport/geometry, and fusion-context score views.
3	Optimize sparse nonnegative component weights on train/CV rows only.
4	Fit a monotone geodesic-odds calibrator and threshold policy without holdout rows.
5	Write `selection_lock.json` and freeze component IDs, weights, calibrator, and threshold source.
6	Score the blind holdout once and emit KPI, calibration, false-alarm, and leakage evidence.

lation/control deltas use the same holdout labels and metric functions as Table VIII. Values copied directly from `selected_component_ablations.csv` are treated as ablation-internal diagnostics unless they share that metric basis.

Table XI is also a guard against over-claiming: the EI claim is the locked calibrated selected-threshold result, not post-hoc dominance over every view control.

VIII. RAW SAMPLES AND COMPARE-ONLY ANCHOR

The raw sample figures are visual diagnostics, not detector features. Figure 8 exposes the model card and signal-chain assumptions, while Figure 9 shows range-Doppler proxy views for a positive public-proxy group and a hard-negative group across the three phase windows.

The compare-only measured anchor is KTH, a 77 GHz FMCW drone/bird/human collection that is useful for distribution-shape checks but not for positive-class truth claims [4]. Figure 10 shows normalized KTH anchor medians and q10/q90 intervals as measured-anchor z-scores, avoiding mixed-unit bars. The purpose is to check whether the synthetic hard-negative observables land in a sensible micro-Doppler and range-style neighborhood, not to promote the public-proxy fixed-wing class into a measured-platform claim; public anchors such as Open Radar, RAD-DAR, and related datasets are used in the same compare-only spirit [5]–[7].

IX. COMPACT DIAGNOSTICS SNAPSHOT

The evidence bundle also contains compact diagnostic summaries that are easier to audit in table form than in figure captions. Table XIII shows the leakage and realism checks, Table XIV shows the selected-threshold confusion matrix, Table XVI shows the EI candidate by phase, and Table XV summarizes the top-method false-positive burden. Table XVII remains a cautious EI false-alarm family diagnostic.

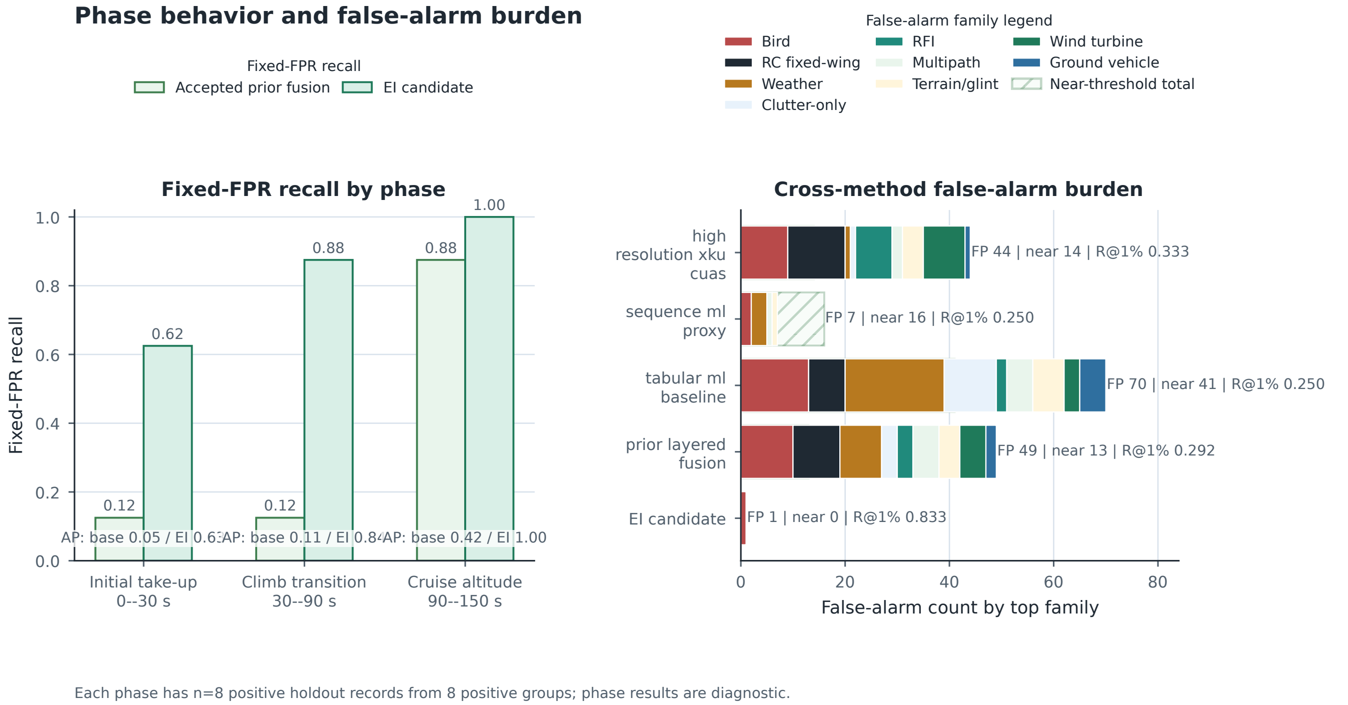

Fig. 5. Per-phase fixed-FPR recall and cross-method false-positive burden. The left panel keeps the operating metric simple: bars are Recall@$\leq$1%FPR for accepted prior fusion and EI, while AP is printed only as diagnostic text. The right-panel false-alarm family legend maps stacked colors to hard-negative families; pale/hatched bars show near-threshold totals, and the FP, near, and R@1% labels are diagnostic robustness context. Each phase has only eight positive holdout records from eight positive groups, so phase conclusions are diagnostic.

TABLE X
READER-FACING EI COMPONENT WEIGHTS SORTED BY WEIGHT

Audit ID	Family	Modality	Calibrator	Weight	Cum.	View
C2	Passive quality	passive-RF provenance	geodesic odds	0.433	0.433	signed DE
C4	Transport geometry	motion/geometry	geodesic odds	0.292	0.725	positive DE
C8	Passive hypergraph	passive-RF graph	geodesic odds	0.153	0.879	signed DE
C1	Passive quality	passive-RF provenance	raw	0.037	0.916	signed DE
C5	Transport geometry	motion/geometry	raw	0.034	0.950	signed DE
C3/C6/C7	Transport/passive residual	mixed	mixed	0.050	1.000	residual

A. *How to Read the Diagnostics*

The raw audit-header check is deliberately run before the model-facing denylist is applied. It confirms that split and group metadata are visible to audit code and blocked before modeling, which is the behavior needed to prove the guard rather than assume it.

The label-shuffle test and the weak metadata baselines provide a separate sanity check. When the labels are permuted, average precision falls from 0.825 to 0.006 and ROC AUC falls from 0.917 to 0.497, which is the collapse expected if the model is responding to detector evidence rather than index structure. The stratum-only and metadata-only baselines are not meant to compete with the detector; they are meant to show that site, range, aspect, and noise metadata alone do not explain the observed ranking performance.

The phase table and the false-alarm tables fill in the operational interpretation. The initial take-up window is still the hardest phase, which is the right failure mode for a radar-oriented benchmark because the target is still transitioning and the observable is least settled. The false alarms are not spread uniformly across every negative family; instead, they concentrate in `single_bird` for the EI candidate and in the bird and RC families for the stronger baseline methods. That is a helpful robustness diagnostic because it identifies where the benchmark can still be stressed without needing any proprietary or operational sensor claims.

Finally, the main experiment lanes, component weights, and comparable view-control rows are audit trails, not claims about a universal ranking of features. They show what the EI meta-fusion candidate uses and keep ablation-internal scores separate from the main holdout-table-compatible metrics. That makes the EI result interpretable in the same way the compare-only KTH overlay is interpretable: as evidence about the benchmark and detector behavior, not as a platform truth claim.

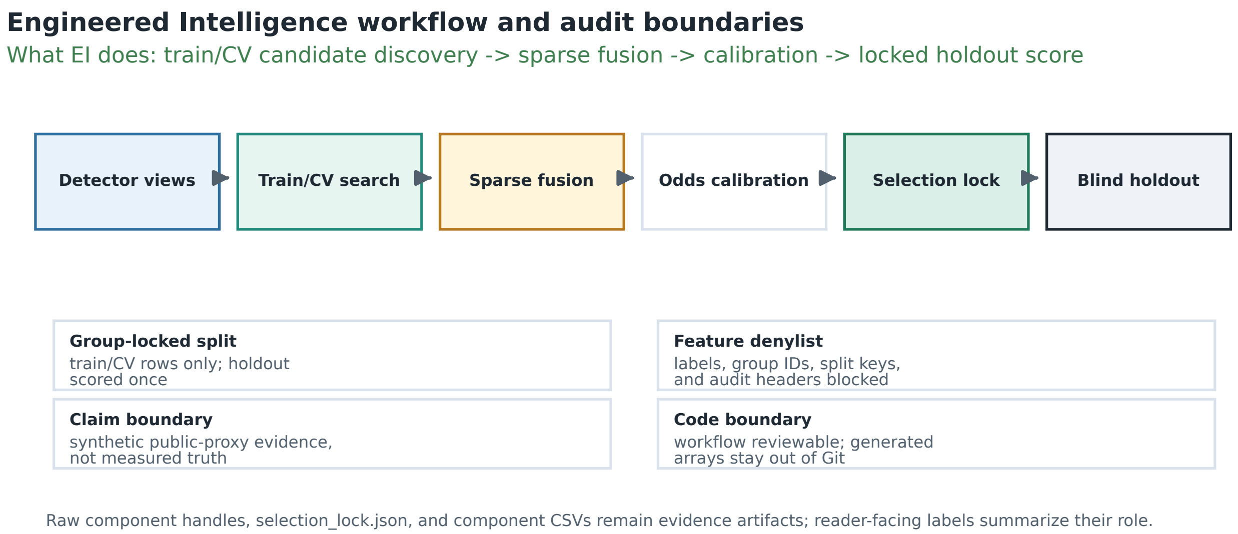

Fig. 6. Engineered Intelligence workflow and audit boundaries. Detector-view tables, train/CV-only candidate discovery, sparse nonnegative fusion, geodesic-odds calibration, EI selection lock, and blind holdout scoring are separated from group-lock, feature-denylist, claim-boundary, and code-disclosure rails.

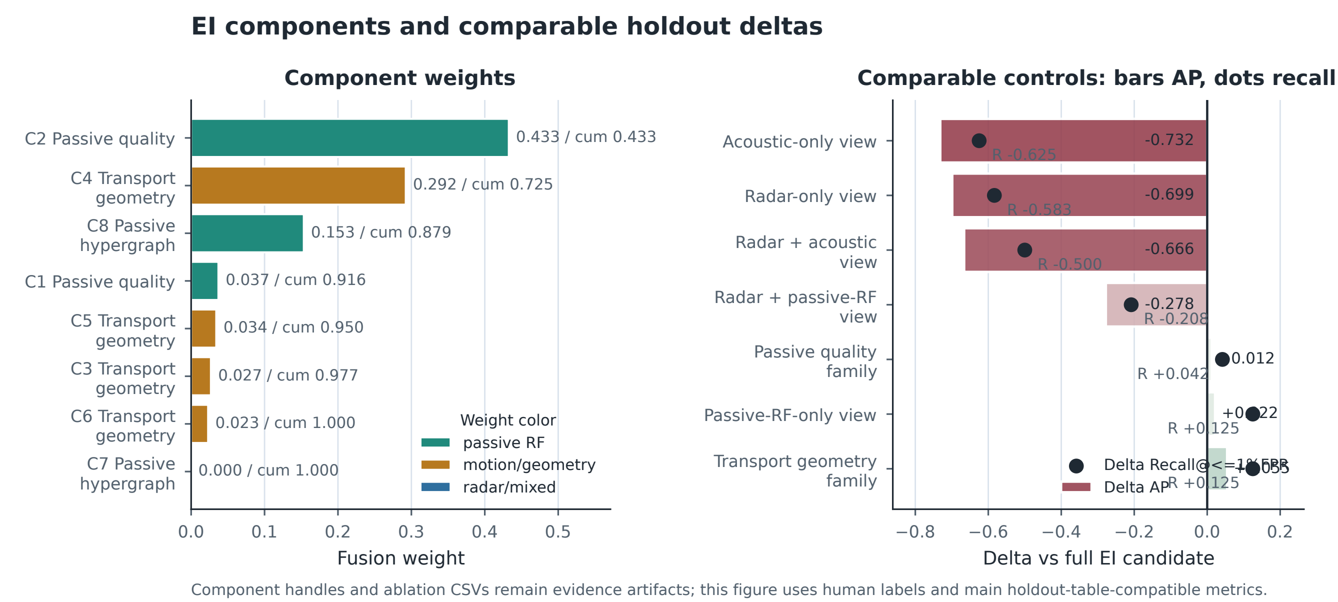

Fig. 7. EI component weights and comparable holdout controls. The left panel sorts human-readable component weights; the right panel reports AP and fixed-FPR recall deltas computed with the same main holdout-table-compatible metric definitions as Table VIII. Raw component handles and ablation-internal diagnostics remain in evidence artifacts.

X. Evidence Ledger

The paper evidence bundle is intentionally split into a small number of human-readable artifacts so that a reviewer can reconstruct the claim boundary without chasing generator internals. The radar model card and detail rows define the public-proxy carrier bands, bandwidths, CPI, PRF/CRF, chirp count, range span, range/Doppler/velocity resolutions, unambiguous-velocity proxy, clutter/noise/RFI priors, RCS prior, and cue definitions. Those are not operational sensor facts; they are explicit synthetic assumptions that make the signal chain legible and make the detector outputs comparable across runs.

The scenario balance files describe the split structure at the scenario-group level, including split, site, range, aspect, weather/noise, hard-negative family, phase, and positive-only balances. That choice matters because a row-level split would allow the same physical group to leak across train and holdout through different phases. By locking the split on groups, the benchmark preserves the structure of the generated motion sequence while still exposing the detector to a proper blind holdout. The balance tables also show that the benchmark is not a single-class toy: the negative families include birds, RC fixed-wing aircraft, weather cells, ground vehicles, wind

TABLE XI
COMPARABLE DETECTOR-VIEW CONTROLS ON THE BLIND HOLDOUT SPLIT

View/control	AP	ΔAP	Recall@≤1%FPR	ΔRecall	F1	ECE
full EI candidate	0.825	0.000	0.833	0.000	0.837	0.002
radar-only view	0.126	-0.699	0.250	-0.583	0.139	0.447
acoustic-only view	0.093	-0.732	0.208	-0.625	0.125	0.754
passive-RF-only view	0.847	+0.022	0.958	+0.125	0.732	0.228
radar+acoustic view	0.159	-0.666	0.333	-0.500	0.225	0.527
radar+passive-RF view	0.547	-0.278	0.625	-0.208	0.526	0.363

Radar model card, resolution budget, and public-proxy boundary

Branch	Band	BW MHz	dR m	PRF/CRF Hz	CPI ms	Pulses	dfD Hz	dv m/s
High-res. X/Ku	X/Ku	600	0.250	4000/4000	6.0	24	166.7	2.498
Tactical S-band	S	180	0.833	4000/4000	6.0	24	166.7	8.059
GBAD 3D/4D	X/Ku cue	300	0.500	4000/4000	6.0	24	166.7	2.426

Public-proxy assumptions

positive class: fixed-wing pusher-prop public proxy
geometry: swept/delta wing; rear pusher propulsor
speed: 45--60 m/s; prop micro-Doppler: 150--217 Hz
phase windows: 0--30 s, 30--90 s, 90--150 s
launch proxy: rail/catapult take-up
RCS/aspect: -28 to -2 dBsm public-source proxy

Receiver impairments

AGC compression	PRF ambiguity
clock drift	Doppler folding
quantization	calibration offset
dropped CPI	multipath masking

Cue definitions

acoustic: node-wise spectral cadence, cross-node agreement, and amplitude stability
passive RF: no-signal / RFI burst / clock-offset / provenance-quality geometry
claim boundary: public-proxy assumptions only

All values are public-proxy or synthetic archetype assumptions; no measured-platform truth is implied.

Fig. 8. Radar model card and signal-chain parameter panel. The figure surfaces the public-proxy assumptions behind the benchmark so the detector results can be read against explicit carrier, bandwidth, PRF/CPI, and cue definitions. The richer positive-proxy and noise/clutter/RFI model cards are in the rich modeling appendix.

TABLE XII
COMPARE-ONLY ANCHOR ROLES

Anchor	EchoForge use boundary
KTH 77 GHz FMCW	Compare-only hard-negative realism and observable-shape checks; no positive truth claim
Open Radar Initiative	Moving-object and micro-Doppler distribution checks after local access rules are satisfied
RAD-DAR/RDRD	Compact range-Doppler comparison after local terms are accepted
Han/Jung 2026	Future multimodal alignment checks, pending license review

turbines, multipath ghosts, RFI bursts, and clutter-only counterfactuals.

The evaluation bundle is equally explicit about how scores are read. Average precision and ROC AUC are reported alongside selected-threshold counts, fixed-FPR recall, group-level operating counts, calibration bins, Brier score, expected calibration error, and bootstrap intervals grouped by scenario. The group-block bootstrap is the right uncertainty model here because the group lock is part of the problem definition; a row bootstrap would overstate certainty by pretending that repeated phases from the same group are independent. This is why the paper reports both aggregate holdout performance and phase-stratified summaries.

The compare-only KTH overlay serves a narrower purpose. It is a measured anchor for observable-shape comparison, useful for checking whether the synthetic hard negatives and the public-proxy positive class live in a plausible neighborhood of micro-Doppler bandwidth, spectral entropy, and range-style observables. The evidence bundle records standardized feature deltas and Wasserstein-style gap rows where local anchor distributions permit them. The anchor does not make the synthetic fixed-wing class into a measured platform proxy, and it should not be read as such.

The EI transparency files complete the ledger by exposing the candidate as an audited composition. The selected compo-

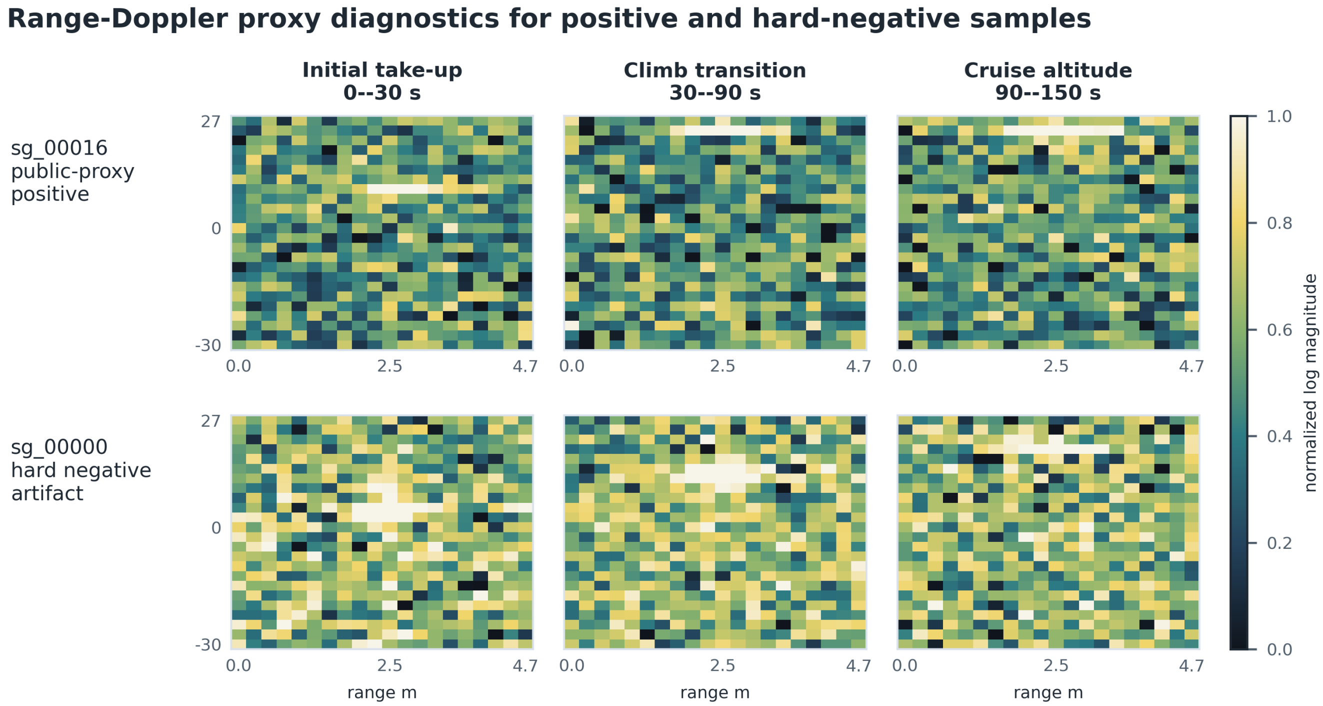

Fig. 9. Qualitative range-Doppler proxy sanity panels for a positive public-proxy group and a hard-negative group. Axes show relative range in meters and velocity-equivalent Doppler bins using the X/Ku branch resolution; color is normalized log-magnitude on a shared scale. These panels are qualitative diagnostics, not measured imagery, not platform-truth evidence, and not the detector input used for the main KPI.

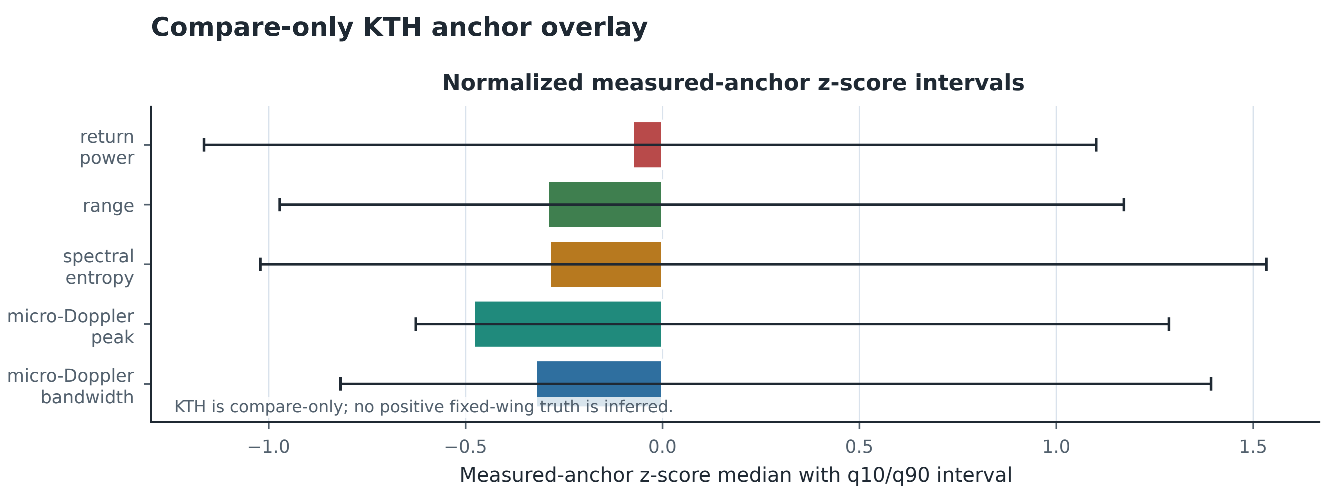

Fig. 10. Compare-only KTH anchor overlay. Features are normalized to measured-anchor z-scores before plotting, so the figure compares distribution shape rather than mixing raw physical units.

nent scores, aliases, human-readable weights, comparable view controls, and quarantined internal ablations make it possible to ask whether a paper figure is using the same metric basis as the main result. In the current run, the EI candidate's false alarms concentrate in a small number of hard-negative families rather than in an undifferentiated mass of failures. That is the kind of evidence a radar reviewer can inspect without being asked to trust a leaderboard jump on faith.

XI. Detector Family Appendix

The evidence bundle includes appendix-ready JSON and CSV cards for each modeled detector/source family. The X/Ku C-UAS card records low-slow-small search, high-resolution range-Doppler, micro-Doppler, clutter suppression, and bird/RC discrimination assumptions. The tactical S-band AESA/MHR card records hemispheric surveillance, track-while-scan confidence, coarser velocity resolution, and multipath stress. The GBAD 3D/4D card records wide-area cueing, revisit delay, radar-horizon masking, and lower-resolution

TABLE XIII
LEAKAGE AND REALISM DIAGNOSTICS

Diagnostic	Value	Interpretation
Raw audit-header detection + detector schema pass	Audit-only headers detected and blocked: `cv_fold`, `scenario_group_id`, `split_role`, and `time_lock_id`	The exported audit views expose these headers for guard testing; the detector-facing schema denylist blocks them before model scoring.
Label-shuffle sanity	EI AP 0.825 vs shuffled AP 0.006; EI ROC AUC 0.917 vs shuffled ROC AUC 0.497; EI ECE 0.002 vs shuffled ECE 0.010	Real labels carry structure; shuffled labels collapse toward chance.
Stratum-only baseline	AP 0.005; ROC AUC 0.457; ECE 0.047	Split metadata alone is weak.
Metadata-only baseline	AP 0.004; ROC AUC 0.473; ECE 0.005	Coarse metadata alone is also weak.
Nearest-neighbor scan	Mean 0.1435, median 0.1226, p10 0.0703, p90 0.2419	Holdout rows are near, but not duplicates of, train rows in the detector-view space.

TABLE XIV
SELECTED-THRESHOLD CONFUSION MATRIX FOR THE PRIOR FUSION BASELINE AND EI CANDIDATE

Method	TP	FP	TN	FN	Precision	Recall	F1	FPR
layered_fusion_c2	10	49	4427	14	0.169	0.417	0.241	0.010947
EI candidate	18	1	4475	6	0.947	0.750	0.837	0.000223

TABLE XV
CROSS-METHOD FALSE-POSITIVE BURDEN FOR THE STRONGEST FIVE METHODS

Method	AP	Recall@≤1%FPR	FP	FP/1000 Neg.	Top FP Family	Burden vs EI
EI candidate	0.825	0.833	1	0.223	single_bird	0
layered_fusion_c2	0.128	0.292	49	10.947	rc_fixed_wing	48
tabular_ml_baseline	0.126	0.250	70	15.639	weather_cell	69
sequence_ml_proxy	0.116	0.250	7	1.564	weather_cell	6
high_resolution_xku_cuas	0.112	0.333	44	9.830	rc_fixed_wing	43

TABLE XVI
EI CANDIDATE BY PHASE ON THE BLIND HOLDOUT SPLIT

Phase	N	Pos	AP	ROC AUC	Recall@≤1%FPR	TP	FP	TN	FN
Initial take-up	1500	8	0.628	0.813	0.625	4	0	1492	4
Climb transition	1500	8	0.845	0.937	0.875	6	1	1491	2
Cruise altitude	1500	8	1.000	1.000	1.000	8	0	1492	0

classification. The classical radar card records CA-CFAR, OS-CFAR, MTD concentration, M/N confirmation, and declared PFA. The acoustic card records spectral cadence, bearing/time correlation, node agreement, weather, traffic, and ambient-noise false cues. The passive-RF card records no-signal behavior, RFI bursts, emitter/provenance cues, and explicit missingness for RF-silent targets. The layered-fusion card records source confidence, stale-track handling, cross-sensor confirmation, and false-track accounting.

XII. SOURCE, NOISE, DETECTOR, AND ASSUMPTION APPENDIX

The appendix ledger separates what each source family is used for from what it is not allowed to prove. This is the reviewer-facing boundary for the public-proxy positive class, hard negatives, detector archetypes, and compare-only measured anchors.

A. Rich Public-Proxy Modeling Appendix

This appendix states how the benchmark models the positive public-source family, the radar/noise environment, the auxiliary cue families, and hard negatives. The public-source

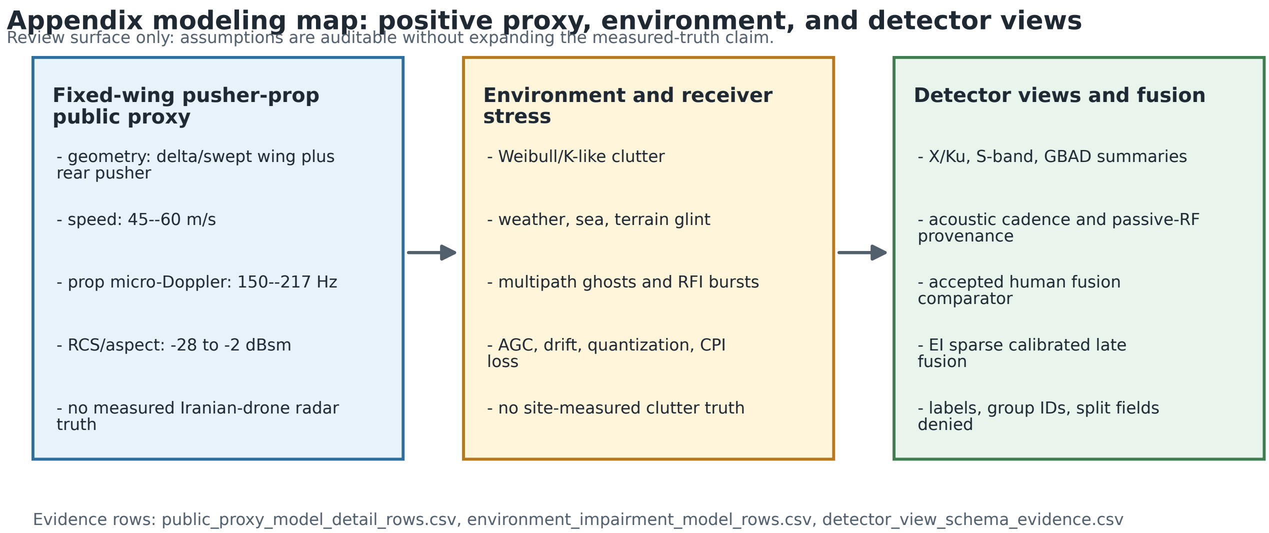

Fig. 11. Appendix modeling map for the positive proxy, environment/receiver stress, and detector views. This figure is a review surface: it shows how assumptions are separated from prohibited measured-truth claims.

TABLE XVII
EI CANDIDATE FALSE-ALARM FAMILIES

Family	Count	FA	Rate	Near
bird_flock	474	0	0.000	0
clutter_only	480	0	0.000	0
ground_vehicle	450	0	0.000	0
multipath_ghost	438	0	0.000	0
rc_fixed_wing	483	0	0.000	0
rfi_burst	387	0	0.000	0
single_bird	405	1	0.002	0
terrain_glint	414	0	0.000	0
weather_cell	486	0	0.000	0
wind_turbine	459	0	0.000	0

positive class is inspired by open reporting on Iranian fixed-wing pusher-prop loitering-munition families, but EchoForge does *not* claim a measured Iranian-drone radar signature, route model, payload model, or operational performance surrogate. The paper-facing class remains the fixed-wing pusher-prop public proxy.

The appendix is not a claim-expansion device. It is a review surface. It explains the synthetic assumptions behind the benchmark while keeping the central claim narrow: EI improves the low-FPR operating point over accepted fusion practice on this declared synthetic public-proxy run.

XIII. Limitations and Prohibited Inferences

The most important limitation is also the simplest: a synthetic benchmark is not measured truth. The present corpus should be read as public-proxy evidence with declared priors, not as a fielded sensor surrogate. The comparison against KTH and other public anchors is a compare-only check, useful for distribution shape, calibration sanity, and hard-negative realism, but insufficient to justify named-platform claims. That boundary follows the usual verification-and-validation distinction in computational science and the reproducibility recommendations for machine-learning research [36]–[39].

The statistical limitation is central. The corpus is a single-seed run with 50 positive groups total and only 8 positive groups, 24 positive records, in the holdout split. Group-block bootstrap intervals are therefore necessary but not sufficient for broad claims. Phase and false-alarm-family slices are robustness diagnostics, not stable estimates of field rates.

The physics limitation is also explicit. The radar branch cards use simplified public-proxy RCS, clutter, weather, RFI, waveform, and processing assumptions. KTH is a 77 GHz measured anchor with a different frequency and target mix, so it can support compare-only observable-shape checks but not measured positive-class validation. The paper contains no hardware-in-the-loop evidence, no measured positive-class validation, no track-level FAR, and no deployment-performance claim. The next useful steps are wider terrain and clutter priors, richer weather and RFI stressors, multi-seed corpora, larger positive holdouts, hardware-in-the-loop hooks, and finer-grained uncertainty reporting.

XIV. Fixed-Wing Pusher-Prop Public-Proxy Appendix

The appendix-only public proxy card records the assumed geometry and flight envelope for the positive class. The boundary is intentionally narrow:

- public-source assumptions: delta or swept fixed wing, rear pusher propulsor, approximate 3.3–3.7 m length, 2.3–2.7 m wingspan, and 0.35–0.75 m height;
- public-source kinematics: 45–60 m/s speed envelope, take-up/climb/cruise phase windows, and a rail/catapult take-up proxy with short booster assist;

TABLE XVIII
FIXED-WING PUSHER-PROP / IRANIAN-DRONE PUBLIC-PROXY MODELING CARD

Modeling item	Public-proxy assumption	Radar consequence	Explicit boundary
Airframe geometry	Delta or swept fixed wing, rear pusher propulsor, approximate 3.3–3.7 m length, 2.3–2.7 m span, 0.35–0.75 m height	Aspect-dependent projected area and RCS envelope	Public visual/summary proxy only; no measured Iranian-platform signature
Launch/take-up	Rail/catapult take-up proxy with short booster-assist stress in the first 0–30 s	Initial phase has ground coupling, multipath, and less settled kinematics	No route, launch-site, operational, or evasion model
Kinematics	45–60 m/s speed envelope across take-up, climb, and cruise windows	Doppler centroid and track-rate priors for branch summaries	Broad public kinematic envelope only
Propulsion micro-Doppler	Pusher-prop cadence proxy of 150–217 Hz	Narrow high-cadence modulation family distinct from many bird wingbeat bands but overlapping RC prop cadence	Not a measured micro-Doppler signature
RCS/aspect	-28 to -2 dBsm aspect envelope with Swerling-like fluctuation proxy	Controls per-branch detectability and hard-negative overlap	Synthetic stress envelope; no sensor-specific Pd/Pfa claim
Phase labels	Initial take-up 0–30 s, climb transition 30–90 s, cruise altitude 90–150 s	Phase-stratified AP, Recall@$\leq$1%FPR, and false-alarm diagnostics	Phase conclusions are diagnostic because the holdout has eight positive groups

TABLE XIX
NOISE, CLUTTER, RFI, AND RECEIVER MODELING DETAILS

Family	What is generated	Why it matters for radar review	Boundary
Thermal/SNR stress	Synthetic SNR/CNR buckets attached to radar branches and detector-view features	Prevents a trivial clean-signal benchmark	Not a calibrated receiver-noise figure for hardware
Weibull/K-like clutter	Terrain, sea, vegetation, and mixed-scene clutter stressors	Covers non-Gaussian clutter regimes used in radar detection literature	Site-measured clutter truth is not claimed
Weather and sea clutter	Rain/dust/weather-cell, coastal haze, and sea/terrain glint families	Creates near-threshold false-track pressure	Not a meteorological rate model
Multipath ghosts	Synthetic ghost tracks and reflection-like interference	Tests whether fusion overreacts to plausible false tracks	No deployment geometry or site-specific ray tracing
RFI bursts and passive-RF missingness	Sparse RF cues, no-signal behavior, burst contamination, clock-offset, and provenance quality	Tests multimodal fusion when RF is absent, sparse, or contaminated	No emitter library, payload command, operator, or exploitation claim
Receiver impairment	AGC compression, clock drift, quantization, dropped CPI, Doppler folding, and calibration offset	Forces detector views to tolerate sensor-like imperfections	Simplified impairment families, not hardware qualification

- public-source observables: an aspect-dependent RCS envelope and a prop micro-Doppler envelope that are only broad proxy bands, not measured signatures.

The appendix uses open visual reporting and public summaries [40]–[42] together with small-UAS radar literature [8], [9]. Prohibited inferences are equally explicit:

- no measured Iranian-drone radar signature;
- no operational route or evasion modelling;
- no payload-effects inference;
- no deployment-performance claim;
- no fielded-sensor equivalence or classified-fidelity claim.

XV. REGIONAL BIRD AND RC HARD-NEGATIVE APPENDIX

The regional bird library is split into hard-negative families that are common around Gulf and coastal sites: single bird, bird flock, shorebird/wader, gull/tern, raptor/falcon, flamingo/large bird, seabird/cormorant, seasonal migratory density, and RC fixed-wing. The evidence bundle records wingbeat bands, body-size/RCS stress notes, site prevalence, seasonality, flock behavior, expected confusion mechanisms, and generated false-alarm summaries in `regional_bird_library.csv` and `source_pack_bird_coverage_summary.csv`.

The source-pack layer includes bird hard-negative cards for the single-bird, dense flock, shorebird/wader, gull/tern, raptor/falcon, flamingo/large-bird, seabird/cormorant, migratory-density, and RC fixed-wing entries, with the pack manifest updated to enumerate those cards explicitly. The compare-only bird literature is public and broad [8], [9], [23]; the point of the library is to expose likely false-positive pressure, not to optimize evasion.

TABLE XX
SYNTHETIC RADAR SIMULATION PROCESSING DETAIL

Processing step	Modeled quantity	How it becomes evidence	Red-line boundary
Target return	aspect-dependent public-proxy RCS, phase-stable complex amplitude, range migration, Doppler centroid	IQ magnitude diagnostics and detector features such as concentration and Doppler spread	no measured platform signature
Micro-Doppler	pusher-prop 150–217 Hz proxy plus bird/RC cadence stressors	branch summaries and hard-negative false-alarm interpretation	cadence bands are broad priors, not identity labels
Clutter/noise	thermal/SNR buckets, Weibull/K-like clutter, sea/weather/terrain glint, vegetation motion	score degradation and near-threshold hard-negative pressure	no site-measured clutter truth
Receiver effects	AGC compression, drift, quantization, dropped CPI, Doppler folding, calibration offset	detector-view robustness features and failure slices	no hardware qualification
Cue streams	acoustic cadence/agreement and passive-RF no-signal/RFI/provenance geometry	non-radar branch scores and EI component inputs	no emitter library or operator inference
Feature contract	denylisted schema converts generated products to detector views	train/CV and holdout scores, leakage diagnostics, and KPI tables	labels, split keys, group IDs, and generator internals blocked

TABLE XXI
DETECTOR-VIEW MODELING CONTRACT

View	Model-facing features	What it can support	What it cannot support
X/Ku radar	Range-Doppler concentration, micro-Doppler spread, CFAR/MTD-style scores, clutter stress	High-resolution radar branch comparison	Proprietary sensitivity, ECCM, or exact vendor Pd/Pfa
S-band radar	Coarser velocity/range summaries, track confidence, multipath stress	Tactical branch stress and cue comparison	Any claimed match to a fielded AESA/MHR implementation
GBAD cueing	Cue confidence, horizon/revisit stress, stale-track state	Wide-area cueing behavior under a public role envelope	Engagement-quality behavior or operator workflow
Acoustic	Spectral cadence, node agreement, weather/traffic stress	Supporting propulsion-cadence cue	Exact microphone layout or field localization performance
Passive RF	No-signal, RFI burst, sparse RF cue, provenance-quality geometry	RF silence/missingness and provenance-quality stress	Emitter identification, command links, payload inference, or operator identity
Fusion	Source confidence, stale-track penalty, cross-sensor confirmation, false-track accounting	Accepted-practice comparator and EI input context	Operational C2 equivalence

XVI. CONCLUSION

EchoForge demonstrates a strict-open path for low-altitude UAS radar and multimodal sensing studies: public-proxy priors, deterministic generation, detector-view denylisting, group-locked holdout evaluation, group-block uncertainty estimates, compare-only measured-anchor checks, and tracked paper evidence. The headline result is LCB95 Recall@$\leq$1%FPR of 0.699 for the EI candidate versus 0.083 for accepted prior fusion, a **+742%** relative gain, with point Recall@$\leq$1%FPR of 0.833 versus 0.292. The benchmark numbers in this paper are synthetic public-proxy evidence under declared assumptions. They are useful for review, robustness work, and reproducible comparison, but they are not measured truth or operational performance claims.

REFERENCES

[1] M. I. Skolnik, *Radar Handbook*, 3rd ed. New York, NY, USA: McGraw-Hill, 2008.

[2] M. A. Richards, *Fundamentals of Radar Signal Processing*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.

[3] H. Rohling, “Radar CFAR thresholding in clutter and multiple target situations,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-19, no. 4, pp. 608–621, Jul. 1983.

[4] A. Karlsson, “Radar measurements on drones, birds and humans with a 77GHz FMCW sensor,” Zenodo, 2021. [Online]. Available: https://zenodo.org/records/5896641

[5] D. Gusland, J. M. Christiansen, B. Torvik, F. Fioranelli, S. Z. Gurbuz, and M. Ritchie, “Open radar initiative: Large scale dataset for benchmarking of micro-doppler recognition algorithms,” in *2021 IEEE Radar Conference (RadarConf21)*. IEEE, 2021.

[6] I. Roldan, C. R. del Blanco, A. Duque Quevedo, F. Ibanez Urzaiz, J. Gismero Menoyo, A. Asensio Lopez, D. Berjon, F. Jaureguizar, and N. Garcia, “Dopplernet: A convolutional neural network for recognising

TABLE XXII
ABLATION AND APPENDIX ARTIFACT MAP

Artifact	Contents	Reviewer question answered
data_processing_trace_rows.csv	Stage-level scenario-to-holdout trace, including input artifacts, output artifacts, reviewer checks, and claim boundaries	Can I reconstruct the processing path without reading generator internals?
main_kpi_gain_table.csv	Prior-fusion and EI values, absolute changes, relative percentages, and gain/reduction direction	What exactly is the +742% LCB95 and +185% point-recall claim computed from?
public_proxy_model_detail_rows.csv	Positive-class geometry, kinematics, RCS, micro-Doppler, launch, phase, and prohibited-inference rows	How is the Iranian-drone public proxy represented without claiming measured truth?
environment_impairment_model_rows.csv	Clutter, noise, weather, RFI, multipath, receiver impairment, and hard-negative stress rows	What noise and artifact regimes are actually modeled?
detector_view_schema_evidence.csv	Allowed feature families and denied audit-only fields for every detector view	Are labels, split keys, group IDs, or generator state leaking into models?
false_alarm_by_method_family.csv	Family-wise false alarms for EI and the strongest comparator methods	Which hard negatives generate false alarms and what do the colors in Fig. 5 mean?
selected_component_human_weights.csv	Reader-facing EI component weights, modalities, calibrators, and cumulative weights	What does EI fuse at a high level?

TABLE XXIII
SOURCE LEDGER FOR PUBLIC-PROXY EVIDENCE

Source family	Used for	Not used for	Evidence artifact
Public fixed-wing pusher-prop reports	geometry, kinematic envelope, qualitative launch-phase priors	measured radar truth, route behavior, payload inference	positive-class card
Small-UAS micro-Doppler literature	broad cadence and spectral-overlap priors	named-platform equivalence or sensor-performance equivalence	radar model card
KTH 77 GHz FMCW	compare-only hard-negative observable-shape checks	positive-class validation or fielded-sensor transfer	normalized anchor comparison
Public detector product pages	detector-branch role envelopes and excluded truth fields	vendor processing, ECCM, or exact Pd/Pfa	sensor archetype cards
Regional bird and RC source pack	hard-negative taxonomy and false-alarm interpretation	evasion optimization or field-rate claims	regional bird library

targets in real scenarios using a persistent range-doppler radar," *IET Radar, Sonar & Navigation*, vol. 14, no. 4, pp. 593–600, 2020.

[7] S.-K. Han and Y.-H. Jung, "A time-synchronized multi-sensor drone dataset acquired from multiple radars and RF receiver," *Scientific Data*, vol. 13, no. 407, 2026.

[8] S. Rahman and D. A. Robertson, "Radar micro-doppler signatures of drones and birds at K-band and W-band," *Scientific Reports*, vol. 8, no. 1, p. 17396, 2018.

[9] P. Molchanov, R. I. A. Harmanny, J. J. M. de Wit, K. Egiazarian, and J. Astola, "Classification of small UAVs and birds by micro-doppler signatures," *International Journal of Microwave and Wireless Technologies*, vol. 6, no. 3–4, pp. 435–444, 2014.

[10] M. Ritchie, F. Fioranelli, H. Borrion, and H. Griffiths, "Multistatic micro-doppler radar feature extraction for classification of unloaded/loaded micro-drones," *IET Radar, Sonar & Navigation*, vol. 11, no. 1, pp. 116–124, 2017.

[11] V. C. Chen, *The Micro-Doppler Effect in Radar*. Norwood, MA, USA: Artech House, 2011.

[12] D. Tahmoush, "Review of micro-doppler signatures," *IET Radar, Sonar & Navigation*, vol. 9, no. 9, pp. 1140–1146, 2015.

[13] M. Ezuma, O. Ozdemir, C. K. Anjinappa, W. A. Gulzar, and I. Guvenc, "Micro-UAV detection with a low-grazing angle millimeter wave radar," in *2019 IEEE Radio and Wireless Symposium (RWS)*. IEEE, 2019, pp. 1–4.

[14] P. Swerling, "Probability of detection for fluctuating targets," *IRE Transactions on Information Theory*, vol. 6, no. 2, pp. 269–308, Apr. 1960.

[15] K. D. Ward, R. J. A. Tough, and S. Watts, *Sea Clutter: Scattering, the K Distribution and Radar Performance*, 2nd ed. London, UK: Institution of Engineering and Technology, 2013.

[16] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume 2: Detection Theory*. Upper Saddle River, NJ, USA: Prentice Hall, 1998.

[17] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part III: Radar-Sonar Signal Processing and Gaussian Signals in Noise*. New York, NY, USA: Wiley, 2001.

[18] Blighter Surveillance Systems, "A400 Series Radars," Product information page, 2026. [Online]. Available: https://blighter.com/products/a400-series-radars/

[19] RTX Raytheon, "KuRFS: Ku-band radio frequency system," Product information page, 2026. [Online]. Available: https://www.rtx.com/raytheon/news/lang/ro/capabilities/products/counter-uas/sensors/kurfs

[20] DRS RADA Technologies, "Next-gen multi-mission hemispheric radar (nMHR)," Product information page, 2026. [Online]. Available: https://www.drsrada.com/products/nmhr

[21] Saab, "Giraffe 1X," Product information page, 2026. [Online]. Available: https://www.saab.com/products/giraffe-1x

[22] RTX Raytheon, "Sentinel radar," Product information page, 2026. [Online]. Available: https://www.rtx.com/raytheon/what-we-do/land/sentinel-radar

[23] Environment Agency – Abu Dhabi, "Birds," Biodiversity information page, 2026. [Online]. Available: https://www.ead.gov.ae/en/Discover-Our-Biodiversity/Birds

[24] G. S. Fishman, *Monte Carlo: Concepts, Algorithms, and Applications*. New York, NY, USA: Springer, 1996.

[25] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, "Equation of state calculations by fast computing machines," *The Journal of Chemical Physics*, vol. 21, no. 6, pp. 1087–1092, 1953.

[26] R. Y. Rubinstein and D. P. Kroese, *Simulation and the Monte Carlo Method*, 3rd ed. Hoboken, NJ, USA: Wiley, 2016.

TABLE XXIV
NOISE, CLUTTER, RFI, AND RECEIVER ASSUMPTION FAMILIES

Family	Benchmark role	Claim boundary
Weibull/K-like clutter	terrain, sea, weather, and clutter-only stressors for radar summaries	synthetic stress envelope, not site-measured clutter truth
RFI burst and passive-RF missingness	sparse cue behavior, RF silence, burst contamination, and provenance quality	no emitter library, command-link, or operator inference
Multipath and ghosting	near-threshold false-track pressure and detector-view robustness	no deployment geometry or site-specific reflection model
Receiver impairments	AGC compression, clock drift, quantization, dropped CPI, Doppler folding, and calibration offset	simplified impairment families, not hardware qualification
Bird and RC hard negatives	low-FPR confusion pressure against public benign aerial families	robustness diagnostics, not evasion guidance or population-rate estimates

[27] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proceedings of the Fourteenth International Joint Conference on Artificial Intelligence*, 1995, pp. 1137–1143.

[28] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.

[29] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proceedings of the 23rd International Conference on Machine Learning*. ACM, 2006, pp. 233–240.

[30] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[31] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*. Cambridge, MA, USA: MIT Press, 1999, pp. 61–74.

[32] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proceedings of the 22nd International Conference on Machine Learning*. ACM, 2005, pp. 625–632.

[33] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[34] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.

[35] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[36] W. L. Oberkampf and C. J. Roy, *Verification and Validation in Scientific Computing*. Cambridge, UK: Cambridge University Press, 2010.

[37] P. J. Roache, *Verification and Validation in Computational Science and Engineering*. Albuquerque, NM, USA: Hermosa Publishers, 1998.

[38] R. D. Peng, "Reproducible research in computational science," *Science*, vol. 334, no. 6060, pp. 1226–1227, 2011.

[39] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Lariviere, A. Beygelzimer, F. d'Alche Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.

[40] Open Source Munitions Portal, "Shahed-131/136 series," Model page, 2026. [Online]. Available: https://osmp.ngo/model/shahed-131-136-series/

[41] Army Recognition, "Shahed-136," Technical data page, 2026. [Online]. Available: https://www.armyrecognition.com/military-products/army/unmanned-systems/unmanned-aerial-vehicles/shahed-136-loitering-munition-kamikaze-suicide-drone-technical-data

[42] Center for Strategic and International Studies, "Putin's missile war: Russia's strike campaign in ukraine," CSIS report, 2023. [Online]. Available: https://csis-website-prod.s3.amazonaws.com/s3fs-public/2023-05/230505_Williams_Putin_Missile.pdf